



## **Part - I: Question and Answer**

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## Long Question

### Introduction

Machine learning systems in production are judged not only by training performance but also by how reliable they remain after deployment. In e-commerce, challenges such as data drift, latency, and model degradation can reduce accuracy, affect user experience, and harm business decisions.

### Challenges

#### 1. Data Drift

Data drift is common in e-commerce because customer behavior, product trends, and seasonal demand can change over time. If a model is trained on older data, its predictions may become less accurate, which can lead to poor recommendations, incorrect demand forecasts, or weaker personalization. A practical solution is to monitor input data and model performance regularly, then retrain the model when major changes are detected. The trade-off is that continuous monitoring and retraining improve reliability, but they also increase engineering effort, compute cost, and maintenance work.

#### 2. Latency

Latency is the time a model takes to return a prediction after receiving an input. In e-commerce, high latency can slow recommendation systems, search ranking, fraud checks, or dynamic pricing, which may reduce user satisfaction and business efficiency. A practical solution is to use lightweight models, optimized serving infrastructure, caching, and batch prediction for tasks that do not require instant output. The trade-off is that making the system faster may require simplifying the model or adding more infrastructure, which can reduce flexibility or increase costs.

#### 3. Monitoring and Model Degradation

Model degradation happens when a deployed model slowly becomes less effective over time, even if it performed well during testing. In an e-commerce setting, this can lead to weaker product recommendations, poor demand prediction, or incorrect customer targeting, which may reduce revenue and user trust. A practical solution is to use continuous monitoring with dashboards, alerts, feedback loops, and regular evaluation of recent data so that performance drops can be detected early. The trade-off is that strong monitoring improves reliability, but it also requires extra infrastructure, engineering effort, and ongoing maintenance.

#### 4. Cross-Functional Collaboration

Reliable ML systems require teamwork between data scientists, engineers and product managers. Data scientists improve model quality, engineers manage deployment and monitoring, and product managers ensure the system supports business goals. This matters because even a strong model can fail if it does not solve the right problem or run reliably in production. Modern tools such as automation, MLOps pipelines, and LLM-based assistants can support monitoring, retraining, documentation, and debugging, but they still need human oversight.

### Short Question : 1

#### Introduction

In machine learning, a good model should not only learn from the training data but also perform well on new unseen data.

#### Overfitting

Overfitting happens when a model learns the training data too closely, including noise and unimportant details. It usually performs very well on training data but poorly on unseen data. For

example, an email classification model may memorize specific words or patterns from the training emails and then fail to classify new emails correctly.

## Underfitting

Underfitting happens when a model is too simple to learn the real patterns in the data. Because of this, it performs poorly on both training data and unseen data. For example, a model used to predict house prices may ignore important factors such as location, size, and condition, so its predictions remain inaccurate even on the training data.

## Generalization and Example

Generalization refers to a model's ability to apply what it has learned from the training data to new unseen data. Both overfitting and underfitting negatively affect generalization. Overfitting reduces generalization because the model depends too much on the training data and does not adapt well to new inputs. Underfitting reduces generalization because the model fails to learn the underlying patterns in the data in the first place.

## Short Question : 2

### FCN vs. Autoencoder:

#### Introduction

Although both FCNs and autoencoders are neural networks, they are designed for different purposes. A fully connected network is used for prediction tasks, while an autoencoder is used to learn a compressed representation of the input and reconstruct it.

#### FCN

A fully connected network has layers in which each neuron is connected to all neurons in the next layer. Its main objective is to take features and predicts an output, such as a class label or a value.

## Autoencoder

An autoencoder has two main parts: an encoder and a decoder. Its purpose is not prediction but learning a smaller representation of the input and then reconstructing the original input as closely as possible.

## Difference, Example, and Application

The main difference is that an FCN usually maps an input to a predicted output, while an autoencoder maps the input back to itself after compression. For example, an FCN can take student marks as input and predict pass or fail, whereas an autoencoder can take a noisy image and reconstruct a cleaner version of it. One common application of autoencoders is image denoising, where the model removes unwanted noise and helps recover the important visual features of the image.